

Inference and linear combinations

Last time: Activity

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-3.989979	1.139951	-3.500	0.000465	***
gre	0.002264	0.001094	2.070	0.038465	*
gpa	0.804038	0.331819	2.423	0.015388	*
factor(rank)2	-0.675443	0.316490	-2.134	0.032829	*
factor(rank)3	-1.340204	0.345306	-3.881	0.000104	***
factor(rank)4	-1.551464	0.417832	-3.713	0.000205	***

Given your friend's GPA (3.95), school (rank 3), and current GRE score (590), what is your estimated probability that they will be accepted to graduate school?

$$\begin{aligned}\hat{\log odds} &= -3.99 + 0.0023(590) + 0.804(3.95) - 1.34 \\ &= -0.797 \\ \hat{\pi}_i &= \frac{e^{-0.797}}{1 + e^{-0.797}} \approx 0.31\end{aligned}$$

Activity

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The standard deviation of GRE scores is 115.5 points. If your friend re-took the GRE and increased their score by one standard deviation, what would be the estimated change in their *odds* of being accepted?

$$e^{0.0023(115.5)}$$

$$\approx 1.3$$

(about a 30% increase in odds)

Activity

estimated probability has improved
from 31% $\rightarrow$ 37%

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If your friend re-took the GRE and increased their score by one standard deviation, what would be their new estimated *probability* of getting accepted?

$$\begin{aligned}\log \text{ odds} &= -3.99 + 0.0023(590 + 115.5) + \\ &\quad 0.804(3.95) - 1.34 \\ &= -0.532 \\ \hat{p}_i &\approx 0.37\end{aligned}$$

95% CI for β_1 :

Activity

$$0.0023 \pm 1.96(0.0011)$$

$$= (0.00014, 0.0045)$$

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Calculate a 95% confidence interval for the change in *odds* of acceptance associated with a *one-unit* increase in GRE score, holding GPA and school rank fixed

$$95\% \text{ CI for } e^{\beta_1} : (e^{0.00014}, e^{0.0045})$$

$$= (1.00014, 1.0045)$$

95% CI for β_1 : (0.00014, 0.0045)

Activity

95% CI for $\beta_1(115,5)$:

Coefficients: (0.0162, 0.520)

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Now calculate a 95% confidence interval for the change in *odds* of acceptance associated with a one *standard deviation* increase in GRE score, holding GPA and school rank fixed

$$\begin{aligned} & 95\% \text{ CI for } e^{\beta_1(115,5)} : (e^{0.0162}, e^{0.520}) \\ & = (1.016, 1.66) \end{aligned}$$

Activity

95% confidence interval for change in *odds* of acceptance associated with a one *standard deviation* increase in GRE score, holding GPA and school rank fixed:

~~(1.014, 1.268)~~ (1.016, 1.66)
i.e. between 1.6% and 66% increase

Your friend is reluctant to retake the exam unless they believe that their odds of acceptance will increase by at least 25%.

Would you recommend that your friend re-takes the exam? ^{probably} No

point estimate: a one-standard deviation increase in GRE score is associated with an increase in odds of 30%

interval estimate: (1.6%, 66%) (a lot of uncertainty)

A new question

So far:

- + Inference for the *change* in log-odds (β_1) or odds (e^{β_1}) for a one-unit increase in X
- + Inference for change in the log-odds and odds for a one *standard deviation* increase in X

New question:

Can we do inference about a *probability*? E.g., can we construct a confidence interval for the true probability of admission, given values of the explanatory variables?

Example

For simplicity, suppose we just look at GRE score for now:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-2.901344	0.606038	-4.787	1.69e-06	***
gre	0.003582	0.000986	3.633	0.00028	***

For a student with a GRE of 590, the *estimated* probability of admission is $\hat{\pi}_i = 0.313$.

We want to construct a confidence interval for the *true* probability of admission, π_i , for this student.

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- + create a confidence interval for the true *log odds* of admission for this student
- + transform the interval into probabilities

Confidence interval for the log odds

For a student with a GRE of 590, the *estimated* log odds is:

$$\widehat{\log \text{ odds}} = \hat{\beta}_0 + \hat{\beta}_1(590)$$

A confidence interval for the *true* log odds is:

$$\widehat{\log \text{ odds}} \pm z^* SE(\widehat{\log \text{ odds}})$$

$$SE(\widehat{\log \text{ odds}}) = \sqrt{\text{Var}(\widehat{\log \text{ odds}})}$$

Variance of a sum of random variables

Fact: For two variables U, V and constants a, b :

$$\text{Var}(aU + bV) = a^2 \text{Var}(U) + b^2 \text{Var}(V) + 2ab \text{Cov}(U, V)$$

where $\underbrace{\text{Cov}(U, V)}_{\text{Covariance}} = \underbrace{\text{Correlation}(U, V)}_{\text{constant}} \cdot \underbrace{\text{SD}(U)}_{\text{Standard deviation}} \cdot \underbrace{\text{SD}(V)}_{\text{Standard deviation}}$

$$\text{Var}\left(\underbrace{\hat{\beta}_0}_{\text{random variables}} + \underbrace{\hat{\beta}_1}_{\text{random variables}} (S_{90})\right) = \text{Var}(\hat{\beta}_0) + (S_{90})^2 \text{Var}(\hat{\beta}_1) + 2(S_{90}) \text{Cov}(\hat{\beta}_0, \hat{\beta}_1)$$

constant

Need:

- $\text{Var}(\hat{\beta}_0)$
- $\text{Var}(\hat{\beta}_1)$
- $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1)$

Finding the variance and covariance

$\swarrow$ variance - covariance
 In R, `vcov` can be used to extract variances and covariances for estimated coefficients:

```

m1 <- glm(admit ~ gre, family = binomial,
           data = grad_app)
vcov(m1)
  
```

```

##              (Intercept)              gre
## (Intercept)  0.3672815851 -5.877568e-04
## gre          -0.0005877568  9.722061e-07
  
```

variance-covariance matrix

$\text{Cov}(\hat{\beta}_0, \hat{\beta}_1)$

$\leftarrow \text{Var}(\hat{\beta}_1)$

$$\begin{aligned}
 \text{Var}(\hat{\beta}_0 + \hat{\beta}_1 (S90)) &= \text{Var}(\hat{\beta}_0) + (S90)^2 \text{Var}(\hat{\beta}_1) + 2(S90) \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) \\
 &= 0.3673 + (S90)^2 (0.0000009722) + 2(S90)(-0.000588) \\
 &= 0.0122
 \end{aligned}$$

$$\Rightarrow SE(\hat{\beta}_0 + \hat{\beta}_1 (S90)) = \sqrt{0.0122} = 0.11$$

Confidence interval for the log odds

For a student with GRE 590, $SE(\widehat{\log \text{ odds}}) = 0.11$

95% confidence interval for log odds:

$$\widehat{\log \text{ odds}} \pm 1.96 SE(\widehat{\log \text{ odds}})$$

$$= -0.788 \pm 1.96 (0.11)$$

$$= (-1.004, -0.572)$$

Confidence interval for the probability

95% confidence interval for the log odds, for a student with GRE 590:

$$(-1.004, -0.572)$$

How would I turn this into a 95% confidence interval for the true *probability* of admission to graduate school?

odds: $(e^{-1.004}, e^{-0.572})$

probability: $\left(\frac{e^{-1.004}}{1 + e^{-1.004}}, \frac{e^{-0.572}}{1 + e^{-0.572}} \right) =$
 $(0.268, 0.361)$

Class activity

Work on the class activity (handout).